

Lesson 1: Parameters and Return Explore

45 minutes

Overview

Students work with envelopes and paper to model functions with parameters and return values. Students create their own physical function envelope for drawing a house that takes in different parameters, and then build another function to calculate and return the cost of building that house.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-3** - Programmers break down problems into smaller and more manageable pieces. By creating procedures and leveraging parameters, programmers generalize processes that can be reused. Procedures allow programmers to draw upon existing code that has already been tested, allowing programmers to write programs more quickly and with more confidence.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

The Pre-Unit Pulse

Warm Up (5 minutes)

Preview Parameterized Functions

Activity (30 minutes)

Functions with Parameters and Return

Wrap Up (10 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Remove specifics from a function so that it can be used in a variety of situations
- Use appropriate vocabulary to describe parameters and return values.

Preparation

- Collect various sizes of sticky notes. Small sticky notes are particularly useful for this activity.
- Two envelopes per group
- Pencils/pens
- One baggy for a teacher demonstration (optional)
- Print out copies of the Activity Guide for students
- Review the Unit 7, Lesson 1 slides prior to the class
- Check the "[Teacher's Lounge](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our [Virtual Lesson Modifications](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [CSP Unit 7 - Lesson 1 - Parameters and Return Explore](#) - Slides

▼ Make a Copy

- **Parameters, Returns, and Libraries** -

Slides [▼ Make a Copy](#)

For the students

- **Activity Guide - Function Houses**
[▼ Make a Copy](#)
- **Functions with Parameters** - Video
([Download](#))
- **Unit 7 Journal (Digital)** - Resource
[▼ Make a Copy](#)
- **Unit 7 Journal (Interactive Notebook)** - Resource
[▼ Make a Copy](#)

Vocabulary

- **Argument** - the value passed to the parameter
- **Parameter** - a variable in a function definition. Used as a placeholder for values that will be passed through the function
- **Return** - used to return the flow of control to the point where the procedure (also known as a function) was called and to return the value of expression

Teaching Guide

The Pre-Unit Pulse

Have your students complete these questions independently some time before starting this unit.

☰ 1-3

The Pre-Unit Pulse

1

2

3

💡 Teaching Tip



Pre-Unit Pulse Questions: The answers to these questions can provide insights into the preferences, strengths, and motivations of your students. If you'd like to adapt the questions or add in your own, you may want to make a copy of [this Google Form](#) which already has the existing questions populated.

For more tips and ideas of how to use these questions, check out the [CSP Guide to Pre-Unit Pulse Questions](#)

Warm Up (5 minutes)

Distribute: Give each student a copy of the Unit 7 Journal ([Digital/Interactive Notebook](#)) or have students set up this unit's section in their CSP journals.

Teaching Tip

Journaling: Journaling can take many different forms, but in general, it's a tool for individual processing and reflection in a form that can be revisited as students develop their skills and understandings. The medium used for journaling can vary depending on classroom needs.

For more guidance and examples, check out the [CSP Guide to Journaling](#)

Preview Parameterized Functions

Remarks

Up to this point, we've learned the basics of programming: variables, conditionals, functions, lists, loops, and traversals. As you grow in your programming skills, you'll start to notice ways that you can organize your code so that it is easier to work with. That's what this unit is all about!

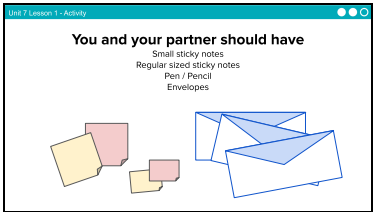
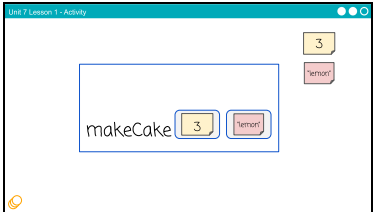
Discuss: *Why would you want to make your code easier to work with or read?*

Discussion Goal: This is a good time to remind students that when we write code, we aren't just writing for a computer - we are also writing for humans who will need to understand our code. Better organization can help others quickly understand our code so they can help debug or build upon the code.

Activity (30 minutes)

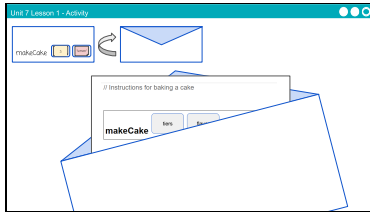
Functions with Parameters and Return

Display: Use the activity slides for this lesson to guide the unplugged activity on Functions with Parameters and Return.

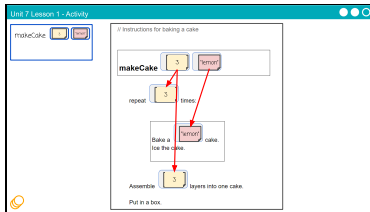
Slides	Speaker Notes
	Distribute: large and small sticky notes and envelopes. Hold off on distributing the Activity Guide until later in the activity.
	Say: Let's pretend that I want to make a cake. I need to decide how many tiers the cake will have, and what flavor the cake will be. I've decided to make a 3 tiered cake.  Click for animation

Slides

Speaker Notes



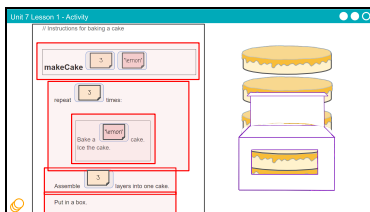
Say: If I flip over the envelope, and open it up there's a recipe inside. This recipe is going to tell me how to make the cake I've specified.



Say: To do this, I'm going to replace `tiers` with `3` and `flavor` with `lemon`, every time those words show up in the recipe. But notice - this recipe is pretty general. I could use it to make lots of different types and sizes of cake.

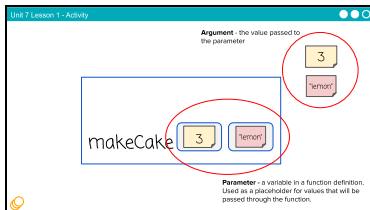
Click for animation

Do This: Read through the recipe as a class. Make sure everyone understands how it works before moving on.



Say: Ok, let's make our cake. Remember, each time I come across `tiers` or `flavor`, I'm going to replace that word with the appropriate value. Once I'm done, I will have baked a 3 layer lemon cake!

Click for animation: There are several steps to click through here. Take it slow so students can see each part.

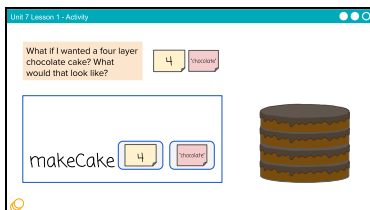


Say: This is a function! It should look familiar, but it has a few extra parts. This function has parameters. Parameters are used as placeholders for values that will be passed through the function.

Click for animation

Say: Those values passed to the parameter are called arguments.

Click for animation



Say: Let's think through how changing the arguments affects the output. Now I want a four layer chocolate cake. With a partner, discuss what my arguments will look like.

Click for animation

Say: I pass through the arguments `4` and `chocolate` to the parameters `tiers` and `flavor`. Then my function is ready. What will my cake look like?

Click for animation

Slides

Speaker Notes

Unit 7 Lesson 1 - Activity

Do This: Time to create your own function with parameters!

- Complete Challenge #1 in your Activity Guide



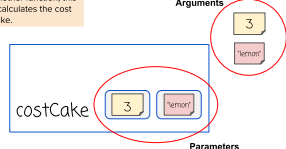
Do This: Distribute Activity Guides and direct students to Challenge #1. Students will create their own function envelopes and write a simple function that uses parameters which can be used to draw a house. When students are finished building their function envelopes, a partner tests out the code and draws the house on a spare sticky note.

Unit 7 Lesson 1 - Activity

Let's look at another function, this time one that calculates the cost of making a cake.

Arguments

Parameters



Say: Here's another function. This function is used to calculate the total cost of making a cake. Take note again of the arguments and parameters.

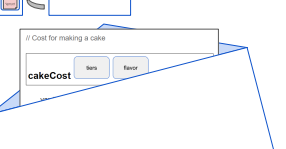
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Unit 7 Lesson 1 - Activity

Let's look at another function, this time one that calculates the cost of making a cake.

Arguments

Parameters



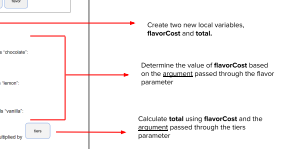
Say: Let's open the envelope and check out how this function works.

Unit 7 Lesson 1 - Activity

Let's look at another function, this time one that calculates the cost of making a cake.

Arguments

Parameters



Say: There are a few new things here. First, we've created two local variables. Remember, the local variables are contained within the function and can't be updated or accessed outside of the function. We learned about this when we discussed variable scope.

 Click for animation

In this function, a decision (in this case, how much to charge per flavor) is made based on the argument passed through the parameter.

 Click for animation

Finally, the total is returned, but returned where? What does that mean?

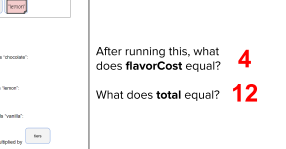
 Click for animation

Unit 7 Lesson 1 - Activity

Let's look at another function, this time one that calculates the cost of making a cake.

Arguments

Parameters



Say: Before we talk more about that return, let's calculate the total cost of our original cake.

Do This: With a partner, determine what `flavorCost` and `total` equal after the function is run.

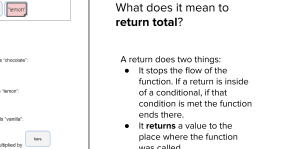
 Click for animation

Unit 7 Lesson 1 - Activity

Let's look at another function, this time one that calculates the cost of making a cake.

Arguments

Parameters



Say: Now let's think about the return. If the program hits a return at any point in running a function, it will stop and a value will be "returned". Where is it returned? To the place in the program where the function was called.

Slides	Speaker Notes
	Say: So to get back to our example, the value 12 is returned, but where is it returned to? Where is it stored? Do This: Take a moment to brainstorm with a partner.
	Say: Let's return to variable baggies! A function return value can be stored in a variable. Notice how the envelope is placed inside of a baggie. That means that the function would be evaluated and whatever is returned would be stored in the variable <code>cakeCalculator</code>. Click for animation
	Say: Here's how this looks in JavaScript. It's ok if it's a little confusing. You'll get to play with this in Code Studio in the next lesson. For now, notice how the statement is set up. A variable is set up with the name <code>cakeCalculator</code>. This gets the value that is returned from the function <code>cakeCost</code> which has two arguments <code>3</code> and <code>lemon</code> passed through the parameters. After the function is evaluated, <code>cakeCalculator</code> now stores the value 12.
	Say: We can also print to the console like so. **Do This: Read through as a class.
	Do This: Direct students to complete Challenge #2 in their Activity Guides. Students create a new envelope function which calculates the cost of the house from Challenge #1. After students are finished, they share the envelopes with a partner who tests the code.

Wrap Up (10 minutes)

Video: Watch the **Functions with Parameters** video as a class. *(Note this video does not cover the concept of functions with return values.)*

Takeaways: Review the slide with the class.

- Functions with parameters and return values help us simplify our code
- Functions can only return one value at a time
- A function can have:
 - No parameters and no return values
 - Parameters, but no return values
 - Return values, but no parameters
 - Parameters and return values

Journal: Students should add the following words and definitions to their journals: parameter, argument, return.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Read through the code and choose the correct answer.



Check For Understanding



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